Inorganic CUME on Materials Chemistry
March 2nd, 2018
Answer any five questions below. Only five will be graded.

1. (20 points) Draw a face-center cubic (fcc) unit cell. How many atoms in a fcc unit cell? In the standard notation of closest-packing of spheres, the letters A, B and C refer to close-packed layers. Which of the following sequences describe the correct closest-packing in 3 dimensions: (i) ABCABC.... (ii) ABABAB…(iii) ABBABABBA....
2. (20 points) Zn metal and many other first row metals readily dissolve in acid and give hydrogen gas. Copper, however, is essentially nonreactive. Rationalize this observation.
3. (20 points) Explain what is (a) Schottky defect, (b) Frenkel defect and (c) Stacking Fault. What are the differences between intrinsic and extrinsic defects?
4. (20 points) A few characterization techniques are mentioned in this paper. Please explain what are a) surface-enhance Raman scattering, b) surface-enhanced fluorescence and c) localized surface plasmon resonance (LSPR) and X-ray photoelectron spectroscopy (XPS).
5. (20 points) What are the first and second law of thermodynamics? What determines the equilibrium shape of a crystal? Why the equilibrium shape of the metal nanocrystal cannot be spherical?
6. (20 points) Explain the roles of capping agents used during synthesis. What are the differences of thermodynamically controlled and kinetically controlled crystal growth? Give an example of each.
7. (20 points) Give the coordination number and charge of the central metal ion in **(a)** $Na_2[Zn(OH)_4]$; **(b)** $K[Co(H_2O)_2(C_2O_4)_2]$; **(c)** $[Ru(H_2O)_2(NH_3)_2Cl_2]Br$.
8. (20 points) For each of the following reactions, use the given rate law to determine the reaction order with respect to each reactant and the overall order.
   (a) $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$; rate = $k[NO]^2[O_2]$
   (b) $CH_3CHO(g) \rightarrow CH_4(g) + CO(g)$; rate = $k[CH_3CHO]^{3/2}$
   (c) $H_2O_2(aq) + 3I^-(aq) + 2H^+(aq) \rightarrow I_3^-(aq) + 2H_2O(l)$; rate = $k[H_2O_2][I^-]$
9. (20 points) For the reaction $N_2O_4(g) = 2NO_2(g)$, $K_c = 0.21$ at 100°C. At a point during the reaction, $[N_2O_4] = 0.12\ M$ and $[NO_2] = 0.55\ M$. Is the reaction at equilibrium? If not, in which direction is it progressing?
10. (20 points) A copper substrate has been covered by a thin film of inorganic nanocrystals. List at least three different characterization techniques that can be combined to study the properties of coating layer including the chemical composition, structure, surface area, thickness and uniformity.